



**Shore**

**Year 12**

**HSC Assessment Task 5 - Trial HSC**

**16<sup>th</sup> August 2013**

# Mathematics Extension 1

## General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this question paper
- Answer Questions 1–10 on the Multiple Choice Answer Sheet provided
- Start each of Questions 11–14 in a new writing booklet
- Show all necessary working in Questions 11–14
- Write your examination number on the front cover of each booklet
- If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A” on the front cover

Examination Number:

Set:

**Total marks – 70**

**Section I** Pages 3–6

**10 marks**

- Attempt Questions 1–10
- Allow about 15 minutes for this section

**Section II** Pages 7–12

**60 marks**

- Attempt Questions 11–14
- Allow about 1 hour 45 minutes for this section

**Note: Any time you have remaining should be spent revising your answers.**

## Section I

**10 marks**

**Attempt Questions 1–10**

**Allow about 15 minutes for this section**

Use the Multiple Choice Answer Sheet for Questions 1–10.

- 1 The point  $P$  divides the interval  $OQ$ , where  $O$  is the origin and  $Q$  is the point  $(3, 7)$ , externally in the ratio  $5 : 4$ . What are the coordinates of  $P$ ?  
(A)  $(-15, -35)$   
(B)  $(-12, -28)$   
(C)  $(12, 28)$   
(D)  $(15, 35)$
- 2 What is the exact value of  $\int_0^{\pi} \cos^2 x \, dx$  ?  
(A)  $0$   
(B)  $\frac{\pi}{4}$   
(C)  $\frac{\pi}{2}$   
(D)  $\pi$
- 3 The function  $f(x) = x^2 - 17$  has a zero near  $x_0$ . Using Newton's Method, which expression gives a better approximation for the zero of the function?  
(A)  $x_0 - \frac{x_0^2 - 17}{2x_0}$   
(B)  $x_0 - \frac{2x_0}{x_0^2 - 17}$   
(C)  $\frac{x_0^2 - 17}{2x_0} - x_0$   
(D)  $\frac{2x_0}{x_0^2 - 17} - x_0$

**DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM**

- 4 The equation  $2x^3 + 5x^2 - 1 = 0$  has roots  $-\frac{1}{2}, \sqrt{2} - 1$  and  $\alpha$ .  
What is the value of  $\alpha$ ?

(A)  $-\sqrt{2} - 1$   
(B)  $-\sqrt{2} + 1$   
(C)  $\sqrt{2} - 1$   
(D)  $\sqrt{2} + 1$

- 5 What is the constant term in the expansion of  $\left(2x^3 - \frac{1}{x}\right)^{12}$  ?

(A)  $-1760$   
(B)  $-220$   
(C)  $220$   
(D)  $1760$

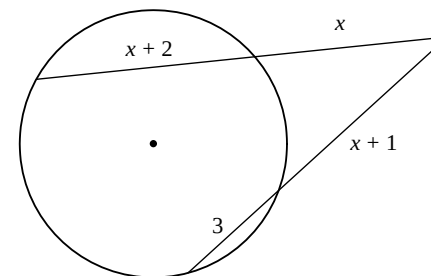
- 6 Which of the following is not a solution of  $2\cos^2 x - 3\cos x + 1 = 0$  ?

(A)  $0$   
(B)  $\frac{\pi}{6}$   
(C)  $\frac{\pi}{3}$   
(D)  $2\pi$

- 7 The line  $y = 2 - x$  and the curve  $y = x^3 + 4$  intersect at the point  $(-1, 3)$ .  
Which of the following is the size in radians of the acute angle between these curves at the point  $(-1, 3)$ ?

(A)  $\tan^{-1} \frac{2}{5}$   
(B)  $\tan^{-1} \frac{1}{2}$   
(C)  $\tan^{-1} 1$   
(D)  $\tan^{-1} 2$

- 8 Two secants from an external point cut off intervals on a circle as shown below.



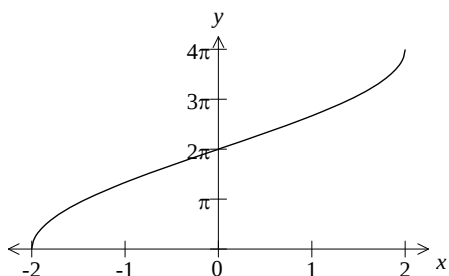
What is the value of  $x$ ?

(A)  $\frac{1 + \sqrt{14}}{2}$   
(B)  $4$   
(C)  $\frac{-3 + \sqrt{73}}{4}$   
(D)  $5$

- 9 A particle is moving in simple harmonic motion according to the equation  $\ddot{x} = -9x + 3$ . Which of the following gives the centre of motion,  $x_0$ , and period,  $T$ ?

- (A)  $x_0 = 3, T = \frac{2\pi}{9}$   
 (B)  $x_0 = 3, T = \frac{2\pi}{3}$   
 (C)  $x_0 = \frac{1}{3}, T = \frac{2\pi}{9}$   
 (D)  $x_0 = \frac{1}{3}, T = \frac{2\pi}{3}$

- 10 The graph below shows a function in the form  $y = a \cos^{-1}(bx)$ .



What are the values of  $a$  and  $b$ ?

- (A)  $a = 4, b = \frac{1}{2}$   
 (B)  $a = \frac{1}{4}, b = -\frac{1}{2}$   
 (C)  $a = 4, b = -\frac{1}{2}$   
 (D)  $a = \frac{1}{4}, b = \frac{1}{2}$

## Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour 45 minutes for this section

Start each of Questions 11–14 in a new writing booklet.

Question 11 (15 marks) Use a SEPARATE writing booklet

Marks

- (a) Solve  $\frac{x+1}{x-1} > 2$ . 3
- (b) Evaluate  $\int_{-3}^3 \frac{1}{\sqrt{9-x^2}} dx$ . 2
- (c) Find  $\frac{d}{dx}(\cos^{-1} e^x)$ . 2
- (d) Use the substitution  $u = x^2 + 2$  to find  $\int_0^1 x(x^2 + 2)^5 dx$ . 3
- (e) Find the exact value of the volume of the solid of revolution formed when the region bounded by the curve  $y = 2 \sec \frac{x}{3}$ , the  $x$ -axis and the lines  $x = 0$  and  $x = \pi$  is rotated about the  $x$ -axis. 3
- (f) Evaluate  $\lim_{x \rightarrow 0} \frac{x^2}{\sin^2 2x}$ . 2

**Question 12 (15 marks)** Use a SEPARATE writing booklet

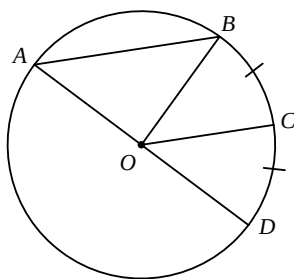
**Marks**

- (a) A roast chicken, which is initially at a temperature of  $220^{\circ}\text{C}$ , is removed from an oven and left to cool on a bench at a constant temperature of  $20^{\circ}\text{C}$ . The cooling rate of the chicken is proportional to the difference between the temperature of the bench and the temperature,  $T$ , of the chicken. That is,  $T$  satisfies the equation

$$\frac{dT}{dt} = -k(T - 20),$$

where  $t$  is the number of minutes after the chicken has been placed on the bench.

- (i) Show that  $T = 20 + Ae^{-kt}$  satisfies this equation. 1
- (ii) The temperature of the chicken is  $60^{\circ}\text{C}$  after 1 hour. Find the temperature of the chicken, to the nearest degree, after 2 hours. 3
- (b) In the diagram below, the points  $A$ ,  $B$ ,  $C$  and  $D$  are concyclic. The point  $O$  is the centre of the circle,  $AD$  is a diameter of the circle and the arc lengths  $BC$  and  $CD$  are equal. Show that  $AB \parallel OC$ . 3



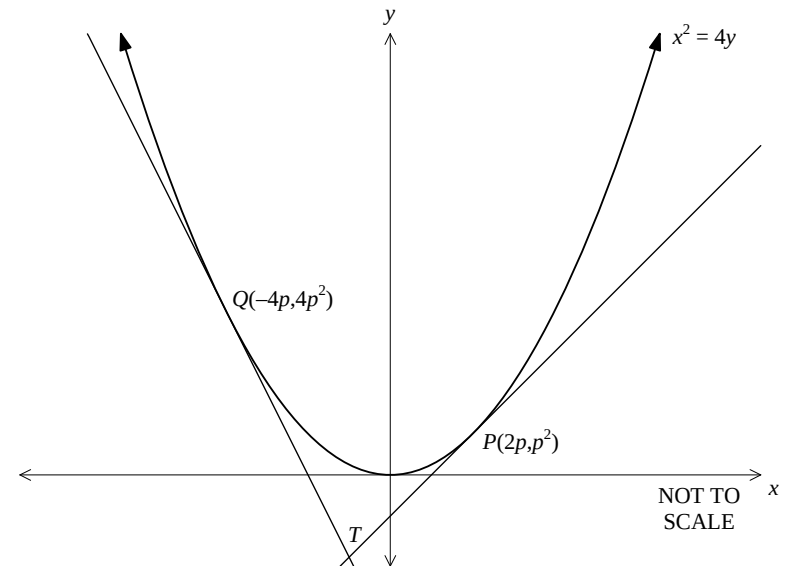
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**Question 12 continues on the following page**

**Question 12 continued**

- (c) Consider the function  $f(x) = \sqrt{x} - 1$ .
- (i) State the domain and range of  $y = f(x)$ . 1
- (ii) Find an expression for  $f^{-1}(x)$ . 2

- (d) The diagram shows the parabola  $x^2 = 4y$ . The points  $P(2p, p^2)$  and  $Q(-4p, 4p^2)$  lie on the parabola. The tangents at  $P$  and  $Q$  intersect at  $T$ .



- (i) Show that the equation of the tangent at  $P$  is given by  $y = px - p^2$ . 2
- (ii) Write down the equation of the tangent at  $Q$ , and find the coordinates of the point  $T$  in terms of  $p$ . 2
- (iii) Find the Cartesian equation of the locus of  $T$ . 1

**Question 13 (15 marks)** Use a SEPARATE writing booklet

**Marks**

- (a) A particle moves in a straight line. Its displacement,  $x$  metres, after  $t$  seconds is given by

$$x = \cos(5t) - \sin(5t).$$

- (i) Show that the particle is moving in simple harmonic motion by showing that  $\ddot{x} = -n^2x$ . 2
- (ii) Express the displacement in the form  $x = R \sin(5t - \alpha)$ , where  $0 < \alpha < \frac{\pi}{2}$ . 2

- (b) A particle moves in a straight line along the  $x$ -axis so that its acceleration is given by  $\ddot{x} = \frac{1}{4 + x^2}$  where  $x$  is the displacement from the origin.

Initially the particle is at rest at the origin.

- (i) Find  $v^2$  as a function of  $x$ . 2
- (ii) Explain why  $v$  is always positive for  $t > 0$ . 1
- (iii) Find the velocity as  $x \rightarrow \infty$ . 1

- (c) Find the greatest coefficient in the expansion of  $(2x + 1)^{18}$ . 3

- (d) The binomial theorem states that 4

$$(1 + x)^n = \sum_{k=0}^n {}^nC_k x^k.$$

By differentiating both sides of this identity twice with respect to  $x$ , show that

$$2^{n-2} - 1 = \sum_{k=2}^{n-1} {}^nC_k \frac{k(k-1)}{n(n-1)}, \quad n \geq 3.$$

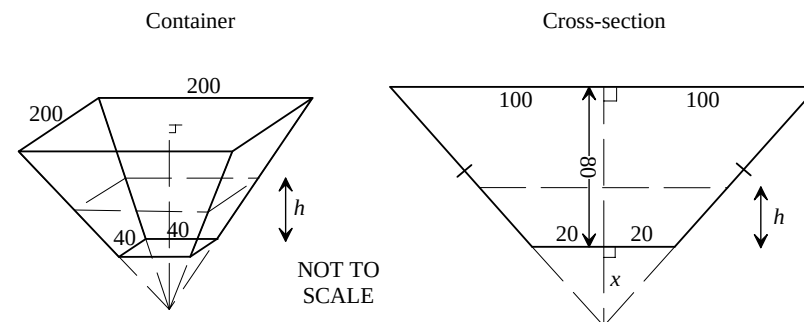
**Question 14 (15 marks)** Use a SEPARATE writing booklet

**Marks**

- (a) Use mathematical induction to prove that 3

$$\sum_{j=2}^n \ln \left( \frac{j-1}{j+1} \right) = \ln \left( \frac{2}{n(n+1)} \right), \quad n \geq 2.$$

- (b) A frustum is a pyramid with its top cut off. Water is being poured into a container in the shape of an inverted right square frustum, as shown below, at a rate of 10 L/s.



The vertical cross-section of the container has the dimensions (in centimetres) shown above.

- (i) Find the value of  $x$ . 1
- (ii) Let the depth of water in the container be  $h$  centimetres. Show that the volume,  $V$ , in  $\text{cm}^3$  of water in the container is given by 2

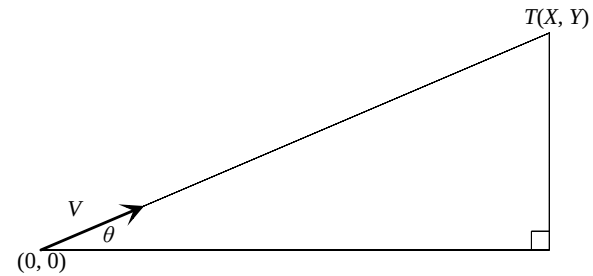
$$V = \frac{4}{3}(h^3 + 60h^2 + 1200h).$$

- (iii) How quickly is the depth increasing when the container has been filled to half its height? 3

**Question 14 continues on the following page**

**Question 14 continued**

- (c) A ball launcher placed at the origin is aimed at a target,  $T$ ,  $X$  metres away horizontally and  $Y$  metres above its launch point, as shown in the diagram below. The initial velocity of the ball is  $V$  m/s.



Assume that at time  $t$  seconds after the launcher is activated the location of the ball is given by

$$\begin{aligned} x &= Vt \cos \theta \\ y &= Vt \sin \theta - 5t^2. \end{aligned}$$

- (i) Show that the equation of the path of the ball is given by 2

$$y = x \left( \frac{Y}{X} \right) - \frac{5x^2}{V^2} \left( 1 + \frac{Y^2}{X^2} \right).$$

- (ii) Find an expression for  $V$  in terms of  $X$  and  $Y$  that will cause the ball to land directly below the target. 2

- (iii) The initial velocity of the launcher is 20 metres per second. 2  
Describe the locus of the target,  $T$ , that will allow the ball to land directly below the target.

**END OF PAPER**

①

## Section I

1. D
2. C
3. A
4. A
5. A
6. B
7. D
8. B
9. D
10. C

$$1. P = \left( \frac{4(0) + -5(3)}{4 + -5}, \frac{4(0) + -5(7)}{4 + -5} \right) \\ = \underline{(15, 35)}.$$

$$2. \int_0^{\pi} \cos^2 x \, dx = \int_0^{\pi} \frac{1}{2} (\cos 2x + 1) \, dx \\ = \frac{1}{2} \left[ \frac{\sin 2x}{2} + x \right] \\ = \frac{1}{2} \left( \left( \frac{\sin 2\pi}{2} + \pi \right) - \left( \frac{\sin 0}{2} + 0 \right) \right) \\ = 0 + \frac{\pi}{2} \\ = \underline{\underline{\frac{\pi}{2}}}.$$

$$3. x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \\ = x_0 - \frac{x_0^2 - 17}{2x_0}$$

$$4. \alpha + \sqrt{2} - 1 - \frac{1}{2} = \frac{-5}{2} \\ \alpha + \sqrt{2} - 1 = -2 \\ \alpha = -\sqrt{2} - 1$$

$$5. \left( 2x^3 - \frac{1}{x} \right)^{12} = \sum_{k=0}^{12} {}^{12}C_k (2x^3)^{12-k} (-x^{-1})^k \\ = \sum_{k=0}^{12} {}^{12}C_k 2^{12-k} x^{36-3k} (-1)^k x^{-k} \\ \text{Constant term when } 36 - 4k = 0 \\ k = 9.$$

②

5. cont.

$$\text{constant term} = {}^{12}C_9 2^3 (-1)^9 \\ = -1760.$$

$$6. 2\cos^2 x - 3\cos x + 1 = 0. \quad \text{let } u = \cos x. \\ 2u^2 - 3u + 1 = 0 \\ (2u-1)(u-1) = 0. \\ \therefore u = 1 \text{ or } u = \frac{1}{2}. \\ \cos x = 1 \text{ or } \cos x = \frac{1}{2} \\ \therefore x = 0, 2\pi \text{ or } x = \frac{\pi}{3}, \frac{5\pi}{3}. \\ \therefore \frac{\pi}{6} \text{ is not a solution}$$

$$7. \text{From } y = 2 - x, \quad m_1 = -1 \\ \text{From } y = x^3 + 4 \\ y = 3x^2 \\ = 3(-1)^2 \text{ when } x = -1. \\ \therefore m_2 = 3. \\ \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\ = \left| \frac{-1 - 3}{1 - 3} \right| \\ = 2 \\ \therefore \theta = \tan^{-1}(2).$$

③

$$8. x(2x+2) = (x+1)(x+4) \\ 2x(x+1) = (x+1)(x+4) \\ (x+1)(2x - (x+4)) = 0 \\ (x+1)(x-4) = 0. \\ \therefore x = 4, -1. \\ \text{But } x > 0 \\ \therefore x = 4.$$

$$9. \ddot{x} = -9x + 3 \\ = -9\left(x - \frac{1}{3}\right) \\ \therefore x_0 = \frac{1}{3}, T = \frac{2\pi}{3}.$$

$$10. a = 4, b = -\frac{1}{2}.$$

④

## Section II

Q11

$$a) \text{ Solve } \frac{x+1}{x-1} = 2, \quad x \neq 1 \\ x+1 = 2x-2 \\ x = 3$$

$\therefore x = 1, x = 3$  are critical points.

$$\begin{array}{c} x \\ | \quad \quad \quad | \\ 1 \quad \quad \quad 3 \end{array} \quad x$$

$\therefore$  solution is  $1 < x < 3$ .

$$b) \int_3^{\frac{3}{2}} \frac{1}{\sqrt{9-x^2}} \, dx = \left[ \sin^{-1} \left( \frac{x}{3} \right) \right]_3^{\frac{3}{2}} \\ = \frac{\pi}{2} - \left( -\frac{\pi}{2} \right) \\ = \pi.$$

$$c) \frac{d}{dx} (\cos^{-1} e^x) = \frac{-1}{\sqrt{1-(e^x)^2}} \cdot e^x \\ = \frac{-e^x}{\sqrt{1-e^{2x}}}.$$

$$d) \int_0^1 x(x^2+2)^5 \, dx = \int_2^3 \frac{u^5}{2} \, du \\ = \left[ \frac{u^6}{12} \right]_2^3 \\ = \frac{3^6}{12} - \frac{2^6}{12} \\ = \frac{665}{12}$$

$$\text{let } u = x^2 + 2 \\ \frac{du}{dx} = 2x \\ x \, dx = \frac{du}{2} \\ \text{When } x = 0, u = 2 \\ \text{When } x = 1, u = 3$$

Q11 cont.

$$\begin{aligned} \text{e) } V &= \pi \int_0^{\pi} 4 \sec^2\left(\frac{x}{3}\right) dx \\ &= 4\pi \left[ 3 \tan\left(\frac{x}{3}\right) \right]_0^{\pi} \\ &= 12\pi (\tan \frac{\pi}{3} - \tan 0) \\ &= 12\pi \sqrt{3} \text{ units}^3 \end{aligned}$$

$$\begin{aligned} \text{f) } \lim_{x \rightarrow 0} \frac{x^2}{\sin^2 2x} &= \lim_{x \rightarrow 0} \left( \frac{2x}{\sin 2x} \right)^2 \times \frac{1}{4} \\ &= 1 \times 1 \times \frac{1}{4} \\ &= \frac{1}{4}. \end{aligned}$$

Q12

$$\begin{aligned} \text{a) (i) } T &= 20 + Ae^{-kt} \quad \Rightarrow T - 20 = Ae^{-kt} \\ \frac{dT}{dt} &= -kAe^{-kt} \\ &= -k(T - 20) \end{aligned}$$

$$\begin{aligned} \text{(ii) When } t &= 60, T = 60^\circ \\ \text{When } t &= 0, T = 20^\circ. \end{aligned}$$

$$\text{So } 220 = 20 + Ae^{-0}$$

$$\therefore A = 200$$

$$\text{So } T = 20 + 200e^{-kt}$$

$$\text{And } 60 = 20 + 200e^{-60k}$$

$$\frac{40}{200} = e^{-60k}$$

$$-\ln 5 = -60k$$

$$k = \frac{\ln 5}{60}$$

Q12 cont.

a) (ii) cont.

$$\begin{aligned} \text{When } t &= 120 \\ T &= 20 + 200e^{-120 \times \frac{\ln 5}{60}} \\ &= 20 + 200e^{-2 \ln 5} \\ &= 28^\circ \text{C}. \end{aligned}$$

$$\begin{aligned} \text{b) Let } \angle COD &= x^\circ \\ \therefore \angle BOC &= x^\circ \quad (\text{equal arcs (BC, CD) subtend equal angles at the centre}) \\ \text{So } \angle BOD &= 2x^\circ \quad (\angle BOC + \angle COD) \\ \therefore \angle BAD &= x^\circ \quad (\text{angle at circumference half angle at centre on same arc}) \\ \therefore \angle ABO &= x^\circ \quad (\text{base angles of isosceles } \triangle ABO, AO = BO, \text{ equal radii}) \end{aligned}$$

$$\text{So } \angle ABO = \angle COB$$

$$\therefore AB \parallel DC \quad (\text{equal alternate angles})$$

$$\text{c) } f(x) = \sqrt{x} - 1$$

$$\begin{aligned} \text{(i) domain: } & x \geq 0 \\ \text{range: } & y \geq -1. \end{aligned}$$

$$\begin{aligned} \text{(ii) } y &= \sqrt{x} - 1 \\ \text{has inverse} \\ x &= \sqrt{y} - 1 \\ x + 1 &= \sqrt{y} \\ y &= (x + 1)^2, \quad x \geq -1. \\ \therefore f^{-1}(x) &= (x + 1)^2, \quad x \geq -1. \end{aligned}$$

Q12 cont.

$$\begin{aligned} \text{d) (i) } y &= \frac{x^2}{4} \\ \frac{dy}{dx} &= \frac{2x}{4} \\ &= \frac{x}{2}. \end{aligned}$$

$$\begin{aligned} \text{Tangent at P: } y - p^2 &= \frac{2p}{2}(x - 2p) \\ y &= px - 2p^2 + p^2 \\ y &= px - p^2 \end{aligned}$$

$$\begin{aligned} \text{(ii) Tangent at Q: } y - (2p)^2 &= -2p(x + 4p) \\ y - 4p^2 &= -2px - 8p^2 \\ y &= -2px - 4p^2 \end{aligned}$$

$$\begin{aligned} \text{At T: } px - p^2 &= -2px - 4p^2 \\ 3px + 3p^2 &= 0 \\ 3p(x + p) &= 0 \\ \therefore x &= -p, \quad y = -2p^2. \end{aligned}$$

$$\text{So } T = (-p, -2p^2).$$

$$\begin{aligned} \text{(iii) } p &= -x \\ y &= -2(-x)^2 \\ y &= -2x^2 \text{ is the locus of T.} \end{aligned}$$

Q13

$$\begin{aligned} \text{a) } x &= \cos 5t - \sin 5t \\ \text{(i) } \dot{x} &= -5 \sin 5t - 5 \cos 5t \\ \ddot{x} &= -25 \cos 5t + 25 \sin 5t \\ &= -25(\cos 5t - \sin 5t) \\ &= -5^2 x \\ \text{which is in the form } \ddot{x} &= -n^2 x, \text{ where } n = 5. \end{aligned}$$

$$\begin{aligned} \text{(ii) Let } \cos 5t - \sin 5t &= R \sin(5t - \alpha) \\ &= R \sin 5t \cos \alpha - R \cos 5t \sin \alpha \\ \therefore -R \sin \alpha &= 1 \\ R \cos \alpha &= -1 \\ \text{So } -\tan \alpha &= -1 \\ \tan \alpha &= 1 \\ \alpha &= \frac{\pi}{4}, \quad (0 \leq \alpha \leq \frac{\pi}{2}). \end{aligned}$$

$$\begin{aligned} \text{Sub. } \alpha &= \frac{\pi}{4} \text{ into } R \cos \alpha = -1 \\ \therefore R &= \frac{-1}{\cos \frac{\pi}{4}} \\ &= \frac{-1}{\frac{1}{\sqrt{2}}} = -\sqrt{2}. \\ \therefore x &= -\sqrt{2} \sin(5t - \frac{\pi}{4}) \end{aligned}$$

$$\text{b) } \ddot{x} = \frac{1}{4+x^2}$$

$$\begin{aligned} \text{(i) } \frac{d(\frac{1}{2}v^2)}{dx} &= \frac{1}{4+x^2} \\ \frac{1}{2}v^2 &= \frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) + C \\ v^2 &= \tan^{-1}\left(\frac{x}{2}\right) + K. \end{aligned}$$

Q13 (cont.)

b) (i) cont.

When  $x=0, v=0$ .

$$\text{So } 0 = \tan^{-1}(0) + K.$$

$$\therefore K = 0.$$

$$\therefore v^2 = \tan^{-1}\left(\frac{x}{2}\right)$$

$$(ii) \ddot{x} = \frac{1}{4+x^2}$$

$> 0$  for all  $x$  as  $x^2 \geq 0$  for all  $x$ .

If the particle begins at rest at the origin and its acceleration is always positive, it will always be moving in the direction of the positive  $x$ -axis.  
i.e.  $v > 0$ .

$$(iii) \lim_{x \rightarrow \infty} v = \lim_{x \rightarrow \infty} \sqrt{\tan^{-1}\left(\frac{x}{2}\right)}$$

$$= \sqrt{\frac{\pi}{2}}.$$

$$c) (2x+1)^{18} = \sum_{k=0}^{18} {}^{18}C_k (2x)^{18-k} 1^k$$

$$= \sum_{k=0}^{18} {}^{18}C_k 2^{18-k} x^{18-k}$$

Q13 (cont.)

c) cont.

$$\text{We want } \frac{{}^{18}C_{k+1} 2^{18-k-1}}{{}^{18}C_k 2^{18-k}} > 1$$

$$\frac{18!}{(18-k-1)!(k+1)!} \times \frac{(18-k)!k!}{18!} \times \frac{1}{2} > 1.$$

$$\frac{18-k}{2(k+1)} > 1$$

$$18-k > 2k+2$$

$$16 > 3k$$

$$k < \frac{16}{3}$$

$\therefore k=5$  is the greatest such integer.

So  ${}^{18}C_6 2^{12} = 76038144$  is the greatest coefficient.

$$d) (1+x)^n = \sum_{k=0}^n {}^nC_k x^k$$

$$\text{Differentiating: } n(1+x)^{n-1} = \sum_{k=0}^n {}^nC_k k x^{k-1}$$

$$\text{Differentiating again: } n(n-1)(1+x)^{n-2} = \sum_{k=0}^n {}^nC_k k(k-1)x^{k-2}$$

$$\text{Let } x=1: n(n-1)2^{n-2} = \sum_{k=0}^n {}^nC_k k(k-1)$$

$$\text{So } 2^{n-2} = 0 + 0 + \sum_{k=2}^{n-1} {}^nC_k \frac{k(k-1)}{n(n-1)} + 1.$$

$$\uparrow \quad \uparrow$$

$$\frac{0 \times -1}{n(n-1)} \quad \frac{1 \times 0}{n(n-1)}$$

$$\uparrow$$

$${}^nC_n \times \frac{n(n-1)}{n(n-1)}$$

$$\therefore 2^{n-2} - 1 = \sum_{k=2}^{n-1} {}^nC_k \frac{k(k-1)}{n(n-1)}$$

Q14

a) Prove true for  $n=2$ .

$$\text{LHS} = \sum_{j=2}^2 \ln\left(\frac{j-1}{j+1}\right) = \ln\left(\frac{1}{3}\right)$$

$$\text{RHS} = \ln\left(\frac{2}{2+3}\right) = \ln\left(\frac{1}{3}\right)$$

LHS = RHS.  $\therefore$  true for  $n=2$ .

Assume true for  $n=k$ .

$$\text{i.e. } \sum_{j=2}^k \ln\left(\frac{j-1}{j+1}\right) = \ln\left(\frac{2}{k(k+1)}\right)$$

Prove true for  $n=k+1$ .

$$\text{i.e. need to show } \sum_{j=2}^{k+1} \ln\left(\frac{j-1}{j+1}\right) = \ln\left(\frac{2}{(k+1)(k+2)}\right)$$

$$\text{LHS} = \sum_{j=2}^{k+1} \ln\left(\frac{j-1}{j+1}\right)$$

$$= \sum_{j=2}^k \ln\left(\frac{j-1}{j+1}\right) + \ln\left(\frac{k}{k+2}\right)$$

$$= \ln\left(\frac{2}{k(k+1)}\right) + \ln\left(\frac{k}{k+2}\right) \quad \text{by the inductive hypothesis.}$$

$$= \ln\left(\frac{2k}{k(k+1)(k+2)}\right)$$

$$= \ln\left(\frac{2}{(k+1)(k+2)}\right)$$

$$= \text{RHS.}$$

$\therefore$  true for  $n=k+1$ .

$\therefore$  by the principle of mathematical induction, true for all  $n \geq 2$ .

Q14 cont.

b) (i) Using similar triangles.

$$\frac{x}{20} = \frac{x+80}{100}$$

$$100x = 20x + 1600$$

$$80x = 1600$$

$$x = 20.$$

(ii)

$$V = \frac{1}{3} A_{\text{top}} h_{\text{top}} - \frac{1}{3} A_{\text{bottom}} h_{\text{bottom}}$$

$$= \frac{1}{3} (2(h+20))^2 (h+20) - \frac{1}{3} (2 \times 20)^2 (20)$$

$$= \frac{4}{3} (h+20)^3 - \frac{4}{3} (20)^3$$

$$= \frac{4}{3} (h^3 + 60h^2 + 1200h + 8000) - \frac{4}{3} (8000)$$

$$= \frac{4}{3} (h^3 + 60h^2 + 1200h)$$

$$(iii) \frac{dV}{dh} = \frac{4}{3} (3h^2 + 120h + 1200)$$

$$\frac{dh}{dt} = \frac{dV}{dt} \times \frac{dh}{dV}$$

$$= 10000 \text{ cm}^3/\text{s} \times \frac{3}{4} \left( \frac{1}{3 \times 40^2 + 120 \times 40 + 1200} \right) \text{ cm/cm}^3$$

$$= 10000 \times \frac{3}{4 \times 10800}$$

$$= 0.694 \text{ cm/s.}$$

Q14

$$c) \text{ (i)} \quad x = vt \cos \theta \\ t = \frac{x}{v \cos \theta}$$

$$\therefore y = v \left( \frac{x}{v \cos \theta} \right) \sin \theta - 5 \left( \frac{x^2}{v^2 \cos^2 \theta} \right)$$

$$= x \tan \theta - \frac{5x^2}{v^2} \left( \frac{1}{\cos^2 \theta} \right)$$

$$= x \left( \frac{y}{x} \right) - \frac{5x^2}{v^2} \left( \frac{1}{\left( \frac{y}{\sqrt{x^2+y^2}} \right)^2} \right)$$

$$= x \left( \frac{y}{x} \right) - \frac{5x^2}{v^2} \left( \frac{v^2}{x^2+y^2} \right)$$

$$y = x \left( \frac{y}{x} \right) - \frac{5x^2}{v^2} \left( 1 + \frac{y^2}{x^2} \right) \quad \blacksquare$$

$$\text{(ii) Directly below target} \Rightarrow x = X, y = 0.$$

$$0 = X \left( \frac{y}{x} \right) - \frac{5x^2}{v^2} \left( 1 + \frac{y^2}{x^2} \right)$$

$$0 = Y - \frac{5}{v^2} (x^2 + y^2)$$

$$\frac{5}{v^2} (x^2 + y^2) = Y$$

$$v^2 = \frac{5}{Y} (x^2 + y^2)$$

$$v = \sqrt{\frac{5}{Y} (x^2 + y^2)}, \text{ as } v > 0.$$

Q14

$$c) \text{ (ii)} \quad 20 = \sqrt{\frac{5}{Y} (x^2 + y^2)}$$

$$400 = \frac{5}{Y} (x^2 + y^2)$$

$$400Y = 5(x^2 + y^2)$$

$$80Y = x^2 + y^2$$

$$x^2 + y^2 - 80Y = 0$$

$$x^2 + y^2 - 80Y + 1600 = 1600$$

$$x^2 + (y - 40)^2 = 40^2$$

$\therefore$  the locus of T is a circle  
centred at (0, 40), radius 40.